
Tools & Technologies Used in V FOODS

A simple guide to the tools, libraries, and services that power V FOODS, explained in plain English.

React 19 + Vite + TypeScript + Tailwind CSS

- **What it is:** The modern frontend technology stack used to build everything you see on the screen.
 - **What it does:**
 - **React 19:** Organizes the screens into modular, lightning-fast components like menu cards, cart drawers, live KDS boards, and order trackers.
 - **Vite 8:** Bundles and serves the code instantly so pages load in less than 500 milliseconds, even on crowded college Wi-Fi or mobile data.
 - **TypeScript:** Enforces strict type contracts and catches bugs during development before any code reaches student or counter tablets.
 - **Tailwind CSS 4:** Modern utility-first styling engine providing clean, responsive layouts, dark modes, and crisp mobile bottom sheets.
 - **TanStack Query (React Query v5):** Manages smart client-side data caching (`stale-while-revalidate`), reducing server read traffic by 80%.
 - **Zustand:** Lightweight global state manager handling the cart, active filters, and live order alerts without bulky boilerplate.
 - **Framer Motion:** Delivers silky micro-animations, slide-over sheets, and smooth interactive feedback.
 - **Why we chose it:** It delivers an instantaneous, native-app-like Progressive Web App (PWA) experience inside mobile browsers without forcing students to download heavy app store binaries.
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Capacitor (Cross-Platform Mobile Shell)

- **What it is:** A bridge tool that wraps the web application into an installable mobile app.
 - **What it does:**
 - Packages the web codebase into installable Android APK and iOS apps.
 - Provides direct hardware access to native device features, such as camera scanning for express counter QR verification and web haptics.
 - **Why we chose it:** Single codebase architecture: one source of truth runs everywhere—on students' iPhones and Android phones, canteen tablets, and desktop admin consoles.
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FastAPI (Python 3.11) + Uvicorn + Pydantic v2

- **What it is:** A high-speed, asynchronous backend engine powering V FOODS API services.
 - **What it does:**
 - Serves as the secure coordination layer between V FOODS, Paytm Payment Gateway, and MSG91 alerts.
 - Generates and verifies cryptographic HMAC-SHA256 signatures for Paytm webhooks and counter QR pickup passes.
 - Features an in-memory catalog cache (`/api/catalog/summary`) that serves 1,000+ simultaneous break-time requests in under 2 milliseconds.
 - Runs background async tasks (via `APScheduler`) to safely cancel expired unpaid pending orders every 5 minutes.
 - Automatically documents and validates all API data models using Pydantic v2.
 - **Why we chose it:** FastAPI handles high concurrent asynchronous I/O with minimal CPU overhead, making it immune to break-time traffic surges while keeping the door open for future Python-based campus food demand forecasting and ML models.
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Supabase

- **What it is:** The secure cloud database and backend platform for V FOODS.
 - **What it does:**
 - Safely holds student wallets, shop menus, and live order queues.
 - Automatically updates kitchen screens instantly the moment an order is placed, with no page refreshing.
 - Runs strict security rules inside the database so users can only view their own data.
 - **Why we chose it:** It gives us bank-grade database security and instant live updates out of the box. Full details are in the `04-the-database.md` guide.
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Paytm Payment Gateway

- **What it is:** The official payment gateway used for wallet top-ups and direct order checkout.
- **What it does:**
 - Lets students load money into their campus wallet using any UPI app (Paytm, Google Pay, PhonePe, BHIM) or bank cards/netbanking.
 - Generates secure cryptographic HMAC-SHA256 checksum signatures to protect every transaction against forgery.
 - Sends a secure, digitally signed webhook confirmation directly from Paytm's servers to our FastAPI backend once payment succeeds.
 - Enables an automated three-way revenue distribution (90% to Canteen, 5% to V Foods Platform, 5% to College) on all completed orders.

- **Why we chose it:** Paytm is widely used across Indian campuses, offers high UPI success rates, provides robust enterprise sub-account split capabilities, and processes webhook confirmations server-to-server so payments never get lost if student Wi-Fi cuts out.
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MSG91

- **What it is:** An enterprise communication service for sending SMS and WhatsApp alerts.
 - **What it does:**
 - Sends an instant SMS and WhatsApp message to the student when their food is marked "Ready for Pickup".
 - Sends one-time passwords (OTP) when users log in.
 - **Why we chose it:** It complies with Indian telecom regulations and delivers messages reliably in seconds, even if a student is sitting in a basement food court with weak mobile data.
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GitHub

- **What it is:** A cloud platform that safely stores the project's code.
 - **What it does:**
 - Tracks every change made to the code over time, acting like an unlimited "undo" history.
 - Lets the team work on different features together without overwriting each other's work.
 - Connects directly to our hosting server so new code updates can be published automatically.
 - **Why we chose it:** It is the industry standard for code storage, team collaboration, and version control.
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Render

- **What it is:** The cloud hosting service that runs V FOODS on the live internet.
 - **What it does:**
 - Keeps the backend server and website running 24/7 at a public web address.
 - Automatically detects when new code is pushed to GitHub, builds the project, and deploys it live with zero downtime.
 - Provides automatic security certificates (HTTPS) so all student data is encrypted.
 - **Why we chose it:** It removes the headache of managing physical servers and keeps the app online automatically.
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Supabase MCP (Model Context Protocol)

- **What it is:** A secure bridge connecting the AI assistant to the live database during development.
- **What it does:**
 - Lets the AI assistant check real table names, column structures, and stored procedures directly.

- Verifies that code and database rules match reality without manual copying and pasting.
 - **Why we chose it:** It prevents guesswork and ensures all code changes match the real, running database.
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Figma MCP (Model Context Protocol)

- **What it is:** A bridge connecting the AI assistant to the original design files.
 - **What it does:**
 - Lets the AI inspect exact colors, spacing, and component designs straight from the design team.
 - Helps make sure the built screens match the approved mockups accurately.
 - **Why we chose it:** It ensures visual consistency and high design fidelity across all screens.
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